

	the same equation as the case for 4(b), so $F = 10 \text{ N}$ as x and w remain the same.
5(a)	Pressure at the bottom of the cylinder is given by $P = \frac{F}{A} \quad \text{where } F = \text{weight of liquid above bottom} = mg = (\rho V)g = \rho Ahg$ Therefore $P = \frac{\rho Ahg}{A} = \rho gh$
5(b)(i)	$V = \pi r^2 h$ $370 \times 10^{-6} = (3.14)(3.0 \times 10^{-2})^2 h$ $h = 0.131 \text{ m} = 0.13 \text{ m}$
5(b)(ii)	$P = 880 \times 9.81 \times 0.131$ $= 1130 \text{ Pa} = 1100 \text{ Pa}$ The newton meter reading measures the tension T of the thread. As the object is in equilibrium, net force acting on the object is zero: $T + U = mg$, where T = tension, U = upthrust due to liquid, mg = weight of object. $T = mg - U$ $T = mg - \rho_l V_o g$, where V_o is the volume of the object, ρ_l is the liquid density $= 3.32 - 880 \times 130 \times 10^{-6} \times 9.81$ $= 2.20 \text{ N} = 2.2 \text{ N}$
5(b)(ii)	
6(a)	Similarity: both quantities are defined as work done per unit charge, hence they